Backend Schema Document

Overview

All data lives in one PostgreSQL database (with pgvector), 24 tables in total. Each module owns its tables and only writes to its own; other modules read them or call the owning module's functions. The patient is the centre: almost every table links back to `patients`.

Data model

Owner	Table	Key columns	Links to
Core	`patients`	patient_id (PK), name, sex, date_of_birth, phone, region, distance_km, opted_out	—
Core	`nhis_membership`	patient_id, nhis_number, valid_from, valid_to	patients
Core	`staff_users`	user_id (PK), name, role (doctor, nurse, front_desk, billing, admin)	—
Core	`audit_log`	id, actor (user or AI), action, entity, entity_id, model, prompt_version, created_at	any
M1	`transcripts`	transcript_id (PK), patient_id, consultation_date, clinic, text, answer_key	patients
M1	`clinical_notes`	id (PK), transcript_id, patient_id, ai_draft, final_note, edited_fields, flags, status, guideline_check, approved_by, approved_at	transcripts, patients, staff_users
M2	`kb_chunks`	chunk_id (PK), document, section, pages, text, embedding, doc_version	—
M2	`rag_queries`	id, source, question, retrieved_chunks, answer, citations, refused, latency_ms	kb_chunks

Owner	Table	Key columns	Links to
M3	`appointments`	appointment_id (PK), patient_id, date, time, visit_type, booked_on, source_note_id, confirmed, rescheduled_from, reminders_sent, attended	patients, clinical_notes
M3	`risk_scores`	appointment_id, score, band, top_factors, explanation, model_version, scored_at	appointments
M4	`messages`	id, patient_id, appointment_id, direction, job, template, body, status, intent, confidence, action, approved_by	patients, appointments
M4	`nurse_inbox`	message_id, status, answered_by	messages
M4	`alerts`	message_id, type, acknowledged_by	messages
M4	`clinic_slots`	date, time, status	appointments
M5	`agent_runs`	id (PK), note_id, patient_id, status, summary, steps_used, tokens, cost_usd, latency_ms	clinical_notes, patients
M5	`agent_steps`	run_id, step_no, reasoning_text, tool_name, tool_input, tool_output, status	agent_runs
M5	`proposed_actions`	id (PK), run_id, type, payload, reason, source, status, decided_by	agent_runs
M5	`plan_flags`	run_id, issue, details, resolved_by	agent_runs
M5	`referral_letters`	id, patient_id, text, created_from_action	patients, proposed_actions
M6	`claims`	claim_id (PK), note_id, patient_id, nhis_number, visit_date, diagnosis_code, total_ghs, status, submission_status, mode	clinical_notes, patients

Owner	Table	Key columns	Links to
M6	`claim_lines`	claim_id, item_type, item_name, quantity, unit_price_ghs, line_total_ghs	claims, tariffs
M6	`claim_flags`	claim_id, check_id, result, message, suggested_fix, citation, override_reason	claims
M6	`nhis_submissions`	claim_id, attempt, decision, reasons, amount_paid_ghs	claims
M6	`tariffs`	item_name (PK), item_type, unit_price_ghs	—

Column details for each module's tables are in that module's architecture doc; this table is the single map of where everything lives.

Key relationships

- One **patient** has many transcripts, notes, appointments, messages and claims.
- One **approved note** creates at most one agent run, one follow-up appointment and one claim. This is the backbone of the product.
- One **agent run** has many steps, proposed actions and flags; each approved action creates exactly one appointment, message, letter or claim.
- One **appointment** has one current risk score and many messages (reminders and replies).
- One **claim** has many lines, flags and submission attempts.

Access control

Who can see (R) or change (W) each area. Enforced on every endpoint by role.

Area	Doctor	Nurse	Front desk	Billing	Admin
Patients (identity, contact)	R	R	R/W	R	R/W
Transcripts and clinical notes	R/W	R	—	R (approved only)	R
Guidelines (ask)	R	R	R	R	R/W (ingest)
Appointments and risk	R	R	R/W (attendance)	—	R

Area	Doctor	Nurse	Front desk	Billing	Admin
Messages, nurse inbox, alerts	R	R/W	R (reminders only)	—	R
Care plans and agent traces	R	R/W (approve)	—	—	R
Claims	—	—	—	R/W (submit)	R
Audit log and results	—	—	—	—	R

Rules every agent must follow: no table stores the masking name map; diagnosis and medicines can only be changed through the Note review screen by a doctor; every write by AI or a person goes to `audit_log`.